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Oscillator frequency based on mobility and PTAT temperature compensation

Abstract

In an example, a system includes a first transistor having a first terminal coupled to a current mirror and a control terminal coupled to a first current source and a resistor. The system includes a second transistor having a first terminal coupled to the current mirror, a second terminal coupled to a second terminal of the first transistor, and a control terminal coupled to the resistor and a second current source. The system includes a third transistor having a first terminal coupled to a voltage terminal, a second terminal coupled to the control terminal of the second transistor, and a control terminal coupled to the first terminal of the second transistor. The system includes a fourth transistor having a control terminal coupled to the current mirror, first and second terminals coupled to one another and to the second terminal of the first transistor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority to India Provisional Patent Application No. 202341058650, which was filed Sep. 1, 2023, is titled “IMPROVED ACCURACY OF OSCILLATOR FREQUENCY BASED ON MOBILITY AND PTAT TEMPERATURE COMPENSATION,” and is hereby incorporated herein by reference in its entirety.

BACKGROUND

(2) Oscillators are electronic circuits that produce periodic signals such as sine waves, square waves, or triangle waves, at a given frequency. Linear or harmonic oscillators produce sinusoidal or near-sinusoidal waves. Relaxation oscillators produce non-sinusoidal output signals, such as square waves, triangle waves, or sawtooth waves. The frequency of many relaxation oscillators is proportional to the resistance and capacitance in the electronic circuit.

SUMMARY

(3) In accordance with at least one example of the description, a system includes a first transistor having a control terminal and first and second terminals, the first terminal coupled to a current mirror and the control terminal coupled to a first current source and a resistor. The system also includes a second transistor having a control terminal and first and second terminals, the first terminal coupled to the current mirror, the second terminal coupled to the second terminal of the first transistor, and the control terminal coupled to the resistor and a second current source. The system includes a third transistor having a control terminal and first and second terminals, the first terminal coupled to a voltage terminal, the second terminal coupled to the control terminal of the second transistor, and the control terminal coupled to the first terminal of the second transistor. The system includes a fourth transistor having a control terminal and first and second terminals, the

control terminal coupled to the current mirror, the first and second terminals coupled to one another and to the second terminal of the first transistor and the second terminal of the second transistor.

(4) In accordance with at least one example of the description, a system includes a first transistor having a control terminal and first and second terminals, the first terminal coupled to a current mirror and the control terminal coupled to a first current source and a resistor. The system also includes a second transistor having a control terminal and first and second terminals, the first terminal coupled to the current mirror, the second terminal coupled to the second terminal of the first transistor, and the control terminal coupled to the resistor and a second current source. The system includes a third transistor having a control terminal and first and second terminals, the first terminal coupled to a voltage terminal, the second terminal coupled to the control terminal of the second transistor, and the control terminal coupled to the first terminal of the second transistor. The system includes a fourth transistor having a control terminal and first and second terminals, the control terminal coupled to the current mirror, the first and second terminals coupled to one another and to the second terminal of the first transistor and the second terminal of the second transistor. The system also includes a comparator having a comparator output and first and second comparator inputs, where the control terminal of the fourth transistor is coupled to the first comparator input.

(5) In accordance with at least one example of the description, a system includes a first transistor having a control terminal and first and second terminals, the first terminal coupled to a current mirror and the control terminal coupled to a first current source and a resistor, the first current source configured to provide a proportional to absolute temperature (PTAT) current. The system also includes a second transistor having a control terminal and first and second terminals, the first terminal coupled to the current mirror, the second terminal coupled to the second terminal of the first transistor, and the control terminal coupled to the resistor and a second current source. The system includes a third transistor having a control terminal and first and second terminals, the first terminal coupled to a voltage terminal, the second terminal coupled to the control terminal of the second transistor, and the control terminal coupled to the first terminal of the second transistor. The system also includes a fourth transistor having a control terminal and first and second terminals, the control terminal coupled to the current mirror, the first and second terminals coupled to one another and to the second terminal of the first transistor and the second terminal of the second transistor, the fourth transistor configured to receive a first current from the current mirror at its control terminal. The system includes a comparator having a comparator output and first and second comparator inputs, where the control terminal of the fourth transistor is coupled to the first comparator input, and where the comparator is configured to compare a voltage at the control terminal of the fourth transistor to a reference voltage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a core of a relaxation oscillator with a frequency dependent on electron mobility in accordance with various examples.
- (2) FIG. 2 is a relaxation oscillator in accordance with various examples.
- (3) FIG. 3 is a graph of waveforms for a relaxation oscillator in accordance with various examples.
- (4) FIG. 4 is a circuit for a relaxation oscillator with a frequency dependent on electron mobility with generic temperature compensation in accordance with various examples.
- (5) FIG. 5 is a method for providing an oscillating output signal in accordance with various examples.
- (6) The same reference numbers or other reference designators are used in the drawings to designate the same or similar (functionally and/or structurally) features.

DETAILED DESCRIPTION

(7) A relaxation oscillator is a non-linear oscillator that produces a non-sinusoidal output signal, such as a triangle wave or a square wave. Relaxation oscillators often use a switching device (such as a transistor or a comparator) that repeatedly charges and discharges a capacitor or inductor through a resistance. The frequency of the oscillator depends on the time constant of the circuit. The time constant characterizes the response of the circuit to a step input. In a resistor-capacitor (RC) based oscillator, the time constant and the frequency of the oscillator are proportional to the values of R and C.

(8) In some semiconductor manufacturing processes, the value of resistances may vary 10-15% from the designed value. The value of capacitances may vary by 10% from the designed value. The combined variances of the resistances and capacitances may result in the variation in frequency of an oscillator by around 25%. To improve the frequency of the oscillator responsive to these variations, trim bits may be useful. However, trim bits use additional area on the chip, trim circuitry, memory cells, and other components. Trim bits also add to testing time and testing cost.

(9) In examples herein, a relaxation oscillator architecture is described, where the frequency of the oscillator is dependent on electron mobility (μ) rather than R and C. Electron mobility is a single parameter rather than two parameters (such as R and C), and is more easily controlled for a given manufacturing process. The examples herein also describe a circuit that can compensate for a variety of temperature coefficients of the manufacturing process. The examples herein show a relaxation oscillator that produces a square wave, but other examples may produce other types of oscillating output signals.

(10) FIG. 1 is a core **100** of a relaxation oscillator with a frequency dependent on electron mobility in accordance with various examples herein. In an example, a circuit provides a current to a transistor configured as a capacitor (e.g. a metal-oxide semiconductor (MOS) capacitor). The temperature variation is cancelled by using a bias current that is proportional to absolute temperature (PTAT) in core **100**. In other examples described below, a divider in the oscillator circuit produces a linear temperature coefficient, which may be compensated by changing a reference voltage of the oscillator proportional to temperature.

(11) Core **100** includes transistors **102**, **104**, **106**, **108**, **110**, **112**, and **114**. Core **100** also includes current sources **116** and **118**, and resistor **120**. Core **100** includes voltage terminals **122** and **124**. Core **100** includes current mirror **126**. A number of currents are also shown in core **100**, such as currents **128**, **130**, **132**, **134**, and **136**.

(12) In core **100**, transistors **102**, **104**, **106**, **108**, **110**, **112**, and **114** are field effect transistors (FETs). The transistors may be metal-oxide semiconductor FETs (MOSFETs) in one example. In other examples, other types of transistors may be useful. In this example, transistors **102**, **104**, **106**, and **108** are n-channel FETs, and transistors **110**, **112**, and **114** are p-channel FETs. For the transistors described herein, a gate terminal may be referred to as a control terminal. The source and drain terminals of the transistor may be referred to herein as a first terminal or a second terminal (or vice versa).

(13) Transistor **102** has a gate, a source, and a drain. The gate (e.g., the control terminal) is coupled to a first terminal of current source **116** and a first terminal of resistor **120**. The drain of transistor **102** (e.g., the first terminal) is coupled to transistor **110** in current mirror **126**. The source (e.g., the second terminal) of transistor **102** is coupled to voltage terminal **124**, current source **118**, the source of transistor **104**, and the source and drain of transistor **108**. Voltage terminal **124** may provide a voltage VSS, which may be any suitable value, including ground.

(14) Transistor **104** has a gate (e.g., the control terminal) coupled to the second terminal of resistor **120**, a first terminal of current source **118**, and a source of transistor **106**. The drain of transistor **104** (e.g., the first terminal) is coupled to the gate of transistor **106** and to transistor **112** in current mirror **126**. The source of transistor **104** (e.g., the second terminal) is coupled to voltage terminal **124**, the second terminal of current source **118**, the source of transistor **102**, and the source and drain of transistor **108**.

- (15) Transistor **106** is configured as a source follower in this example. Transistor **106** has a gate (e.g., the control terminal) coupled to the drain of transistor **104** and transistor **112** in current mirror **126**. The drain of transistor **106** (e.g., the first terminal) is coupled to voltage terminal **122**. The source of transistor **106** (e.g., the second terminal) is coupled to the first terminal of current source **118**.
- (16) Transistor **108** has a gate (e.g., the control terminal) coupled to the drain of transistor **114** in current mirror **126**. Transistor **108** has a drain (e.g., a first terminal) and a source (e.g., a second terminal) coupled to one another and coupled to voltage terminal **124**. Transistor **108** is configured as a MOS capacitor. One terminal of the MOS capacitor is the gate of transistor **108**, and the other terminal is the source and drain of transistor **108**. The capacitance of the MOS capacitor depends on the voltage at the gate terminal. The source and drain of transistor **108** may be coupled to ground in one example.
- (17) Transistors **110**, **112**, and **114** make up current mirror **126**. Transistor **110** has a gate (e.g., a control terminal) coupled to the gate of transistor **112** and to the drain (e.g., the first terminal) of transistor **110**. Transistor **110** has a source (e.g., a second terminal) coupled to voltage terminal **122**. Transistor **112** has a gate (e.g., a control terminal) coupled to the gates of transistors **110** and **114**. Transistor **112** has a drain (e.g., a first terminal) coupled to the gate of transistor **106** and the drain of transistor **104**. Transistor **112** has a source (e.g., a second terminal) coupled to voltage terminal **122**. Transistor **114** has a gate (e.g., a control terminal) coupled to the gate of transistor **112**. Transistor **114** has a drain (e.g., a first terminal) coupled to the gate of transistor **108**. Transistor **114** has a source (e.g., a second terminal) coupled to voltage terminal **122**.
- (18) First current source **116** has a first terminal coupled to the gate of transistor **102** and resistor **120**. First current source **116** has a second terminal coupled to voltage terminal **122**. Second current source has a first terminal coupled to the source of transistor **106** and a second terminal coupled to voltage terminal **124**. Resistor **120** has a first terminal coupled to the gate of transistor **102** and a second terminal coupled to the gate of transistor **104**.
- (19) In an example herein, voltage terminal provides a voltage $V_{sub,DD}$, and voltage terminal **124** provides a voltage $V_{sub,SS}$. $V_{sub,SS}$ may be ground in one example. First current source **116** provides a current **134** that is a PTAT current ($I_{sub,PTAT}$). Second current **118** provides a current **136**. Current **136** may be larger than current **134**, such as at least twice as large in one example. In one example, transistor **110**, **112**, and **114** are approximately the same size. In an example, transistor **104** has a width to length ratio approximately four times the width to length ratio of transistor **102**.
- (20) In an example operation, current mirror provides currents **128**, **130**, and **132**. These currents are similar in size and may have a value of $I_{sub,1}$. Current sources **116** and **118** provide bias currents that bias transistors **102** and **104**, and current source **118** provides bias current for transistor **106**. Current **132** is provided to transistor **108**, the MOS capacitor. The voltage at the gate of transistor **108** is provided to a comparator (not shown in FIG. 1), and switches (not shown in FIG. 1) charge and discharge the MOS capacitor to provide an oscillating output signal. The switches and comparator and their operation are described below with respect to FIG. 2.
- (21) Core **100** is able to produce an oscillating signal with a frequency based on electron mobility (μ) based on the following equations. In these equations, transistor **102** is referred to as MN1, transistor **104** as MN2, transistor **110** as MP1, and transistor **112** as MP2. The current through transistor MP2 (transistor **112**) is current **130** in FIG. 1. This current is defined in Equation (1):
- (22)
$$I_{MP2} = \frac{1}{2} C_{OX} \frac{4W_{MN1}}{L_{MN1}} (V_{gs2} - V_{th})^2 \quad (1)$$
- (23) In Equation (1), μ is the electron mobility of the process, $c_{sub,OX}$ is the oxide capacitance of the process, $W_{sub,MN1}$ is the width of transistor **102** (MN1), $L_{sub,MN1}$ is the length of transistor **102**, $V_{sub,gs2}$ is the gate to source voltage of transistor **104** (MN2), and $V_{sub,th}$ is the threshold voltage of transistor **104**. As noted above, the W/L ratio of transistor **104** is four times the W/L ratio

of transistor **102** in this example, so $4W_{\text{sub.MN1}}=W_{\text{sub.MN2}}$, and $L_{\text{sub.MN1}}=L_{\text{sub.MN2}}$.

(24) Equation (2) is the current through MP1 (transistor **110**):

$$(25) \quad I_{MP1} = \frac{1}{2} C_{OX} \frac{W_{MN1}}{L_{MN1}} (V_{gs2} + I_{PTAT} R - V_{th})^2 \quad (2)$$

(26) As seen in Equation (2), the voltage values in parentheses now include the $I_{\text{sub.PTAT}}$ Current (current **134**) times the resistance R, which is the resistance of resistor **120**. Current mirror **126** provides that currents **128** and **130** are equal, and therefore $I_{\text{sub.MP1}}=I_{\text{sub.MP2}}$. The difference between the gate to source voltages of transistor **102** and transistor **104** is $I_{\text{sub.PTAT}}R$. Because $I_{\text{sub.MP1}}=I_{\text{sub.MP2}}$, Equation (2) may be rewritten as Equation (3):

$$(27) \quad I_{MP2} = \frac{1}{2} C_{OX} \frac{4W_{MN1}}{L_{MN1}} (I_{PTAT} R)^2 \quad (3)$$

(28) As described above, transistor **108** is configured as a MOS capacitor. The capacitance value of the MOS capacitor may be referred to herein as $C_{\text{sub.1}}$. The capacitance value $C_{\text{sub.1}}$ is given by Equation (4):

$$(29) \quad C_1 = C_{OX} (W_{C1} * L_{C1}) \quad (4)$$

(30) In Equation (4), $c_{\text{sub.OX}}$ is the oxide capacitance of the process, $W_{\text{sub.C1}}$ is the width of the MOS capacitor (transistor **108**), and $L_{\text{sub.C1}}$ is the length of the MOS capacitor. The delta voltage (e.g., the change in voltage) $\Delta V_{\text{sub.C1}}$ across the MOS capacitor caused by the current **132** flowing into the MOS capacitor is given by Equation (5):

(31) $V_{C1} = \frac{I_{MP1}}{C_1} * t$ (5) where $I_{\text{sub.MP1}}$ is the value of current **132**, $C_{\text{sub.1}}$ is the capacitance value from Equation (4), and t is time. Substituting for $I_{\text{sub.MP1}}$ and $C_{\text{sub.1}}$ in Equation (5) produces Equation (6):

$$(32) \quad V_{C1} = \frac{\frac{2 * W_{MN1}}{L_{MN1}} (I_{PTAT} R_1)^2}{W_{C1} L_{C1}} * t \quad (6) \quad \text{where } R_{\text{sub.1}} \text{ is the value of resistor } \mathbf{120}. \text{ Rewriting}$$

Equation (6) to solve for frequency f (e.g., $1/t$), yields Equation (7):

$$(33) \quad f = V_{C1} * \frac{W_{C1} L_{C1}}{\frac{2 * W_{MN1}}{L_{MN1}} (I_{PTAT} R_1)^2} \quad (7)$$

(34) In Equation (7), frequency f is dependent on electron mobility and bias current $I_{\text{sub.PTAT}}$, which both have temperature variation. The temperature variation may be cancelled by using a temperature proportional bias current ($I_{\text{sub.PTAT}}$), as shown in the following equations. First, the example manufacturing process described herein has an electron mobility of about 2.2. Therefore, the electron mobility and its relationship to temperature is shown in Equation (8):

$$(35) \quad \mu = \mu_0 * T^{-2.2} \quad (8) \quad \text{where } \mu_{\text{sub.o}} \text{ is the standard electron mobility and } T \text{ is temperature.}$$

The $I_{\text{sub.PTAT}}$ current is shown in Equation (9):

(36) $I_{PTAT} = \frac{V_{BE}}{R}$ (9) where $\Delta V_{\text{sub.BE}}$ is proportional to temperature T . Also, it follows from Equation (9) that $(\Delta V_{\text{sub.BE}})^{\text{sup.2}}$ is proportional to temperature $T^{\text{sup.2}}$. From Equation (6), if electron mobility μ is proportional to $T^{\text{sup.-2.2}}$, and $I_{\text{sub.PTAT}}^{\text{sup.2}}$ is proportional to $T^{\text{sup.2}}$ (via $\Delta V_{\text{sub.BE}}$), then $V_{\text{sub.C1}}$ is proportional to $T^{\text{sup.-0.2}}$. Therefore, temperature cancellation is achieved. The temperature variation of electron mobility and bias current $I_{\text{sub.PTAT}}$ approximately cancel each other out. Equation (6) may be rewritten as Equation (10):

$$(37) \quad V_{C1} = \frac{\frac{2 * W_{MN1}}{L_{MN1}} (\frac{V_{BE}}{R} * R_1)^2}{W_{C1} L_{C1}} * t \quad (10)$$

(38) Equation (7) may be rewritten as Equation (11):

$$(39) \quad f = V_{C1} * \frac{W_{C1} L_{C1}}{\frac{2 * W_{MN1}}{L_{MN1}} (\frac{V_{BE}}{R} * R_1)^2} \quad (11)$$

(40) The process described above provides temperature cancellation if the manufacturing process has a temperature coefficient of about 2 (e.g., about 2.2 in one example). However, other processes may have different temperature coefficients. If a process has a temperature coefficient that differs from 2, an alternative circuit may be useful for providing temperature compensation. One

alternative circuit that compensates for a variety of temperature coefficients is described below with respect to FIG. 4.

(41) FIG. 2 is a relaxation oscillator **200** in accordance with various examples herein. Relaxation oscillator **200** includes many of the components of core **100** described above, and also includes additional components such as a comparator and switches to produce an oscillating output. The oscillating output in this example is a square wave that can be programmed to have a variety of duty cycles. Other types of oscillating outputs may be produced in other examples.

(42) Relaxation oscillator **200** includes transistors **102**, **104**, **106**, **108**, **110**, and **112**. Relaxation oscillator **200** also includes current sources **116** and **118**, and resistor **120**. Relaxation oscillator **200** includes voltage terminals **122** and **124**. A number of currents are also shown in relaxation oscillator **200**, such as currents **128**, **130**, **134**, and **136**.

(43) Relaxation oscillator **200** also includes transistors **202**, **204**, **206**, and **208**. In this example, transistors **202** and **204** are p-channel FETs, and transistors **206** and **208** are n-channel FETs. Relaxation oscillator **200** includes switches **210** and **212**. Relaxation oscillator **200** includes current mirror **214**. Relaxation oscillator **200** includes comparator **216** that has first comparator input **218A**, second comparator input **218B**, and comparator output **220**. Relaxation oscillator **200** includes inverter **222**, which has inverter input **224** and inverter output **226**. Relaxation oscillator **200** also includes switches **228** and **230**, voltage terminals **232** and **234** (e.g., voltage reference terminals), and currents **236**, **238**, and **240**.

(44) The components of relaxation oscillator **200** that are found in core **100** are configured and operate as described above with respect to FIG. 1. These components include transistors **102**, **104**, **106**, **108**, **110**, and **112**, current sources **116** and **118**, resistor **120**, voltage terminals **122** and **124**, and currents **128**, **130**, **134**, and **136**. The additional components of relaxation oscillator **200** are described below.

(45) Transistor **202** has a control terminal (e.g., a gate) coupled to the control terminal of transistors **112** and **204**, a first terminal (e.g., a drain) coupled to a first terminal of switch **210**, and a second terminal (e.g., a source) couple to voltage terminal **122**. Transistor **204** has a control terminal (e.g., a gate) coupled to the control terminal of transistor **202**, a first terminal (e.g., a drain) coupled to a first terminal of transistor **208**, and a second terminal (e.g., a source) couple to voltage terminal **122**. Transistors **110**, **112**, **202**, and **204** make up current mirror **214**. In one example, transistors **110** and **112** are approximately the same size. Transistor **204** may be approximately four times as large as transistor **202** in one example.

(46) Transistor **206** has a control terminal (e.g., a gate) coupled to the control terminal of transistor **208**. Transistor **206** has a first terminal (e.g., a drain) coupled to a first terminal of switch **212**. Transistor **206** has a second terminal (e.g., a source) coupled to voltage terminal **124** and a second terminal of transistor **208**.

(47) Transistor **208** has a control terminal (e.g., a gate) coupled to its first terminal (e.g., a drain) and to the control terminal of transistor **206**. Transistor **208** has a first terminal coupled to its control terminal and to the first terminal of transistor **204**. Transistor **208** has a second terminal (e.g., a source) coupled to voltage terminal **124** and a second terminal of transistor **206**. Transistors **206** and **208** are configured as a current mirror in one example.

(48) Switch **210** has a first terminal coupled to the drain of transistor **202** and a second terminal coupled to the gate of transistor **108**. Switch **212** has a first terminal coupled to the drain of transistor **206** and a second terminal coupled to the gate of transistor **108**. The gate of transistor **108** is also coupled to comparator **216** (e.g., the second comparator input **218B** of comparator **216**).

(49) Comparator **216** has a first comparator input **218A**, a second comparator input **218B**, and a comparator output **220**. First comparator input **218A** is coupled to switches **228** and **230**. Second comparator input **218B** is coupled to the gate of transistor **108** and switches **210** and **212**. Comparator output **220** is coupled to inverter input **224**.

(50) Inverter **222** has an inverter input **224** and an inverter output **226**. Comparator **216** produces an

OUT signal at comparator output **220**. The OUT signal is received by inverter **222** at inverter input **224**. Inverter **222** produces an OUT_BAR signal at inverter output **226**. The OUT and OUT_BAR signals are inverse from one another. The OUT and OUT_BAR signals are used for switching switches **210**, **212**, **228**, and **230** as described below.

(51) Switch **228** has a first terminal coupled to first comparator input **218A** and a second terminal coupled to voltage terminal **232**. Voltage terminal **232** provides a voltage $V_{sub.REF_HIGH}$. Switch **230** has a first terminal coupled to first comparator input **218A** and a second terminal coupled to voltage terminal **234**. Voltage terminal **234** provides a voltage $V_{sub.REF_LOW}$. The voltages $V_{sub.REF_HIGH}$ and $V_{sub.REF_LOW}$ provide the duty cycle of the relaxation oscillator **200** as described below.

(52) In operation, comparator **216** toggles the charging and discharging currents provided to transistor **108** as the voltage at the gate of transistor **108** reaches $V_{sub.REF_HIGH}$ or $V_{sub.REF_LOW}$. First, the OUT_BAR signal closes switches **210** and **228**. The OUT signal opens switches **212** and **320**. Current mirror **214** provides a current **236** to the gate of transistor **108** if switch **210** is closed. Current **236** may be similar in size to currents **128** and **130** in one example. Transistor **108** is a MOS capacitor, so current **236** charges the capacitor. If switch **228** is closed, the voltage value $V_{sub.REF_HIGH}$ is provided to first comparator input **218A** via switch **228**. Because $V_{sub.REF_HIGH}$ is a higher value than the charge at the gate of transistor **108** at this time (while transistor **108** is charging), comparator output **220** provides $V_{sub.REF_HIGH}$ at its Output. $V_{sub.REF_HIGH}$ is a static high voltage value, while the voltage value at second comparator input **218B** is a lower but rising value.

(53) After current **236** charges the voltage value at the gate of transistor **108**, this voltage value provided to second comparator input **218B** becomes greater than $V_{sub.REF_HIGH}$, causing comparator output **220** to switch. If comparator output **220** switches, the switches controlled by OUT close (switches **212** and **230**), and the switches controlled by OUT_BAR open (switches **210** and **228**). With switch **230** closed and switch **228** open, the voltage $V_{sub.REF_LOW}$ is provided to first comparator input **218A**. With switch **212** closed and switch **210** open, transistor **108** stops charging and begins discharging. Current **238** through switch **212** discharges the voltage at the gate of transistor **108**. Current **240** flows through transistor **208** towards voltage terminal **124**, and the current mirror of transistors **206** and **208** also pull current **238** from the gate of transistor **108** through switch **212** and towards voltage terminal **124**. Current **238** therefore discharges the MOS capacitor of transistor **108**.

(54) During discharge, the voltage at the gate of transistor **108** drops. This voltage is provided to second comparator input **218B**, and eventually drops to a value lower than $V_{sub.REF_LOW}$, which is provided to first comparator input **218A**. If the voltage at the gate of transistor **108** drops below $V_{sub.REF_LOW}$, the comparator output **220** switches. If comparator output **220** switches, switches **212** and **230** open, and switches **210** and **228** close. With switch **210** closed, the voltage at the gate of transistor **108** begins charging again via current **236**. With switch **228** closed, the voltage $V_{sub.REF_HIGH}$ is provided to first comparator input **218A**. Therefore, the charging and discharging cycle repeats, and the switching occurs at the voltage values set by $V_{sub.REF_HIGH}$ and $V_{sub.REF_LOW}$.

(55) The voltages $V_{sub.REF_HIGH}$ and $V_{sub.REF_LOW}$ may be set to any suitable values to provide the high and low values for the oscillating output signal of the relaxation oscillator **200**. The oscillating output signal is provided at comparator output **220**. The voltage at the gate of transistor **108** charges until it reaches $V_{sub.REF_HIGH}$, and discharges until it reaches $V_{sub.REF_LOW}$. The ratio of the sizes of transistor **202** and **204** may be selected to provide a given duty cycle for the oscillating output signal. Current **236** charges transistor **108**, and current **240** discharges transistor **108**. Therefore, if current **240** is four times as large as current **236**, transistor **108** will discharge four times as fast as it charges, to provide a duty cycle of 80%. If, in another example, transistors **202** and **204** are equal in size, then currents **236** and **240** are

approximately equal in size. In this example, transistor **108** charges at about the same rate that it discharges, providing a duty cycle of 50%. Example voltage waveforms are described below with respect to FIG. 3.

(56) FIG. 3 is a graph **300** of waveforms for a relaxation oscillator in accordance with various examples herein. The relaxation oscillator may be relaxation oscillator **200** in one example. Graph **300** includes two waveforms, waveform **302** and waveform **304**. In graph **300**, the x-axis is time in milliseconds, and the y-axis is voltage in volts.

(57) Waveform **302** is the voltage at the gate of transistor **108**, referred to here as $V_{sub.C1}$. This is the voltage that is provided to second comparator input **218B**. $V_{sub.C1}$ is a voltage that is charged and discharged via switches **210** and **212** as described above. In this example, $V_{sub.C1}$ charges to about 3.0 V, and discharges to about 1.0 V. $V_{sub.REF_HIGH}$ is set to 3.0 V, and $V_{sub.REF_LOW}$ is set to 1.0 V. Waveform **302** shows that $V_{sub.C1}$ charges more slowly than it discharges. As described above, the relative lengths of time of the charge and discharge cycles provides the duty cycle for the oscillating output signal. A larger charging time creates a higher duty cycle in this example. The example in FIG. 3 has a duty cycle of about of 80%, so the charging time is four times as long as the discharging time. The charging and discharging times may be set by the relative current values of currents **236** and **240**. In FIG. 2, current **240** is approximately four times as large as current **236**, due to transistor **204** being approximately four times as large as transistor **202**. Therefore, the duty cycle is 4:1, or 80%.

(58) Waveform **304** is the voltage $V_{sub.OUT}$ at the output **220** of comparator **216** in one example (e.g., the OUT signal). Comparator **216** produces the OUT signal that extends between approximately 5.0 V and 0.0 V. The OUT signal has an 80% duty cycle in this example. As seen in waveform **304**, $V_{sub.OUT}$ is high while $V_{sub.C1}$ is charging, and $V_{sub.OUT}$ is low while $V_{sub.C1}$ is discharging. As $V_{sub.C1}$ charges, comparator **216** produces a high output signal at 5.0 V. The signal at first comparator input **218A** is higher than the signal at second comparator input **218B** as $V_{sub.C1}$ is charging. As $V_{sub.C1}$ discharges, comparator **216** produces a low output signal at 0.0 V. The signal at first comparator input **218A** is lower than the signal at second comparator input **218B** as $V_{sub.C1}$ is discharging. Therefore, the charging and discharging cycles of $V_{sub.C1}$ shown in waveform **302** produce the oscillating output signal shown in waveform **304**.

(59) FIG. 4 is a circuit **400** for a relaxation oscillator with a frequency dependent on electron mobility with generic temperature compensation in accordance with various examples herein. In an example, circuit **400** provides a current to a MOS capacitor. Circuit **400** can compensate for a process electron mobility μ that varies with any temperature coefficient. A divider in the oscillator circuit produces a linear temperature coefficient, which may be compensated by changing a reference voltage of the oscillator.

(60) Circuit **400** includes many of the components of core **100** described above, and also includes components to compensate for a general temperature coefficient X, rather than a temperature coefficient of about 2 or 2.2 in core **100**. Circuit **400** includes transistors **102**, **104**, **106**, **108**, **110**, and **112**. Circuit **400** includes resistor **120** and voltage terminals **122** and **124**. Circuit **400** also includes transistors **402**, **404**, **406**, **408**, and **410**. Circuit **400** includes current sources **412** and **414**, and divider **416**. Circuit **400** also includes a number of currents **418**, **420**, **422**, **424**, and **426**.

(61) In this example, transistors **102**, **104**, **106**, and **108** are n-channel FETs, and transistors **110**, **112**, **402**, **404**, and **406** are p-channel FETs. Transistors **408** and **410** are bipolar-junction transistors (BJTs) in this example. The components of relaxation oscillator **200** that are found in core **100** are configured and operate as described above with respect to FIG. 1. These components include transistors **102**, **104**, **106**, **108**, **110**, and **112**, resistor **120**, and voltage terminals **122** and **124**. The additional components of circuit **400** are described below.

(62) Circuit **400** includes transistor **402**. Transistor **402** has a control terminal (e.g., a gate) coupled to the gate of transistor **112**. Transistor **402** has a first terminal (e.g., a drain) coupled to a first terminal (e.g., a collector) of transistor **408**. Transistor **402** has a second terminal (e.g., a source)

coupled to voltage terminal **122**.

(63) Transistor **404** has a control terminal (e.g., a gate) coupled to the gate of transistor **406** and to the first terminal (e.g., a drain) of transistor **404**. The first terminal of transistor **404** is also coupled to a first terminal (e.g., a collector) of transistor **410**. Transistor **404** has a second terminal (e.g., a source) coupled to voltage terminal **122**.

(64) Transistor **406** has a control terminal (e.g., a gate) coupled to the gate of transistor **404**.

Transistor **406** has a first terminal (e.g., a drain) coupled to the gate of transistor **108**. Transistor **406** has a second terminal (e.g., a source) coupled to voltage terminal **122**.

(65) Transistor **408** has a base (e.g., a control terminal) coupled to its first terminal (e.g., a collector) and to the input of divider **416**. Transistor **408** has a first terminal coupled to the first terminal of transistor **402**. Transistor **408** has a second terminal (e.g., an emitter) coupled to voltage terminal **124**.

(66) Transistor **410** has a base (e.g., a control terminal) coupled to the output of divider **416**.

Transistor **410** has a first terminal (e.g., a collector) coupled to the first terminal of transistor **404**.

Transistor **410** has a second terminal (e.g., an emitter) coupled to voltage terminal **124**.

(67) Current source **412** has a first terminal coupled to voltage terminal **122**, and has a second terminal coupled to the gate of transistor **102** and to resistor **120**. Current source **414** has a first terminal coupled to the gate of transistor **104**, resistor **120**, and the source of transistor **106**. Divider **416** has an input coupled to the base of transistor **408** and an output coupled to the base of transistor **410**.

(68) Current **418** is the current (I.sub.1) through transistor **102**. Current **420** is the current (I.sub.OSCO) through transistor **104**. Current **422** is the current (I.sub.OSC) through transistor **408**. Current **424** is the current (I.sub.C2) through transistor **410**. Current **426** is the current (I.sub.C1) provided to the gate of transistor **108** that charges the MOS capacitor of transistor **108**.

(69) In one example, the width to length ratios of transistors **110**, **112**, **402**, **404**, and **406** are approximately equal. The width to length ratio of transistor **104** is four times the width to length ratio of transistor **102** in one example.

(70) In one example operation, current source **412** produces an I.sub.ZTC (zero temperature coefficient) current. In this example, I.sub.ZTC is a bandgap reference current, as shown in Equation (12):

(71) $I_{ZTC} = \frac{V_{BG}}{R}$ (12) where V.sub.BG is a bandgap voltage reference. Current source **414**

produces a current approximately twice that of current source **412** in one example. Therefore, the current from current source **414** is shown in Equation (13):

(72) $2 * I_{ZTC} = 2 * \frac{V_{BG}}{R}$ (13)

(73) In FIG. 4, current **420** is labeled I.sub.OSCO. Current **420** is produced by the current mirror of transistors **110** and **112**. Current **422** is I.sub.OSC, and is shown in Equation (14):

(74) $I_{OSC} = I_{OSCO} * T^X$ (14) where X is the temperature exponent of the given manufacturing process. With the use of current sources **412** and **414**, the current **422** (I.sub.OSC) is also shown in Equation (15):

(75) $I_{OSC} = 2 * \mu_o * T^X * (C_{OX} * \frac{W_{n1}}{L_{n1}} (I_{ZTC} * R))^2$ (15) where μ is electron mobility, X is the temperature exponent of the process, C.sub.OX is the oxide capacitance of the process, W.sub.n1 is the width of transistor **102**, L.sub.n1 is the length of transistor **102**, and I.sub.ZTC*R=V.sub.BG (from Equation (12) above). Therefore, I.sub.OSC is defined by a V.sub.BG/R current rather than an I.sub.PTAT current as described above with respect to FIG. 1.

(76) In FIG. 4, the base to emitter voltage (V.sub.BE1) of transistor **408** is defined by Equation (16):

(77) $V_{BE1} = X * V_t * \ln((I_{OSCO} * T) / I_S)$ (16) where X is the known temperature exponent of

the process, $V_{sub.t}$ is the thermal voltage of transistor **408**, and $I_{sub.S}$ is the saturation current of transistor **408**. Because divider **416** divides by X , the base to emitter voltage ($V_{sub.BE2}$) of transistor **410** is $V_{sub.BE1}$ divided by X , as shown in Equation (17):

(78) $V_{BE2} = V_t * \ln((I_{OSCO} * T) / I_S)$ (17) where $V_{sub.t}$ is the thermal voltage of transistor **410**, and $I_{sub.S}$ is the saturation current of transistor **410**. Transistors **408** and **410** are the same size in one example, and their thermal voltages and saturation currents may also be similar. Divider **416** therefore removes the temperature exponent X from $V_{sub.BE2}$. With a known temperature exponent X , divider **416** may be designed or programmed to divide by the appropriate value of X . Because divider **416** removes the temperature exponent X from $V_{sub.BE2}$, the current through $V_{sub.BE2}$ (e.g., current **424** $I_{sub.C2}$) no longer has a temperature exponent X . Current **422**, as shown above in Equation (15), includes a TX component. Current **424** ($I_{sub.C2}$) is current **422** without the TX component, as shown in Equation (18):

$$(79) \quad I_{C2} = 2 * \frac{1}{\alpha} * T * (C_{OX} * \frac{W_{n1}}{L_{n1}} (I_{ZTC} * R))^2 \quad (18)$$

(80) Therefore, temperature exponent X for a process may be compensated for with circuit **400**.

(81) The current mirror of transistors **404** and **406** provide the current $I_{sub.C2}$ (current **424**) to the gate of transistor **108**, which is shown in FIG. 4 as current **426** ($I_{sub.C1}$). The change in voltage $\Delta V_{sub.C1}(T)$ at the gate of transistor **108** is provided by Equation (19):

$$(82) \quad V_{C1}(T) = \frac{2 * \frac{1}{\alpha} * T * \frac{W_{n1}}{L_{n1}} (I_{ZTC} * R)^2}{W_{C1} * L_{C1}} \quad (19) \quad \text{where } W_{sub.C1} \text{ is the width of transistor } \mathbf{108}, \text{ and } L_{sub.C1} \text{ is the length of transistor } \mathbf{108}.$$

(83) FIG. 5 is a method **500** for providing an oscillating output signal in accordance with various examples herein. The steps of method **500** may be performed in any suitable order. The hardware components described above with respect to FIG. 2 may perform method **500** in some examples. Any suitable hardware, software, or digital logic may perform method **500** in some examples.

(84) Method **500** begins at **510**, where a capacitor is charged with a first current. As an example, current **236** charges transistor **108**, which is configured as a MOS capacitor.

(85) Method **500** continues at **520**, where a voltage at the capacitor is provided to a first input of a comparator. In one example, the voltage at the gate of transistor **108** is provided to the second input **218B** of comparator **216**.

(86) Method **500** continues at **530**, where the comparator compares the voltage at the capacitor to a first reference voltage. The voltage from the MOS capacitor is compared to the voltage at first input **218A** of comparator **216**. At this time, the voltage $V_{sub.REF_HIGH}$ is provided to first input **218A**.

(87) Method **500** continues at **540**, where the comparator switches its output responsive to the voltage at the capacitor exceeding the first reference voltage. As the MOS capacitor charges, the voltage at the second input **218B** of comparator **216** increases, until it exceeds the voltage $V_{sub.REF_HIGH}$. At that time, the output of the capacitor switches, and the OUT and OUT_BAR signals also switch. In one example, switches **210** and **228** open, and switches **212** and **230** close.

(88) Method **500** continues at **550**, where the capacitor is discharged with a second current. In one example, the second current is current **238**, which flows across the closed switch **212** towards transistor **206** and then to voltage terminal **124**.

(89) Method **500** continues at **560**, where the voltage at the gate of the transistor **108** is again provided to the second input **218B** of comparator **216**.

(90) Method **500** continues at **570**, where the comparator compares the voltage at the gate of transistor **108** to a second reference voltage. At this time, the voltage $V_{sub.REF_LOW}$ is provided to first input **218A** of comparator **216**.

(91) Method **500** continues at **580**, where the comparator switches its output responsive to the voltage at the capacitor falling below the second reference voltage. As the MOS capacitor discharges, the voltage at the second input **218B** of comparator **216** decreases, until it falls below

the voltage V.sub.REF_LOW. After that time, the output of the comparator 216 switches again.

(92) Although not shown herein, a processor (or controller) may be configured to read and execute computer-readable instructions to perform the examples described herein. For example, the processor may be configured to invoke and execute instructions stored in a memory. In some examples, the processor may be implemented as a generic or custom processor or controller capable of executing instructions stored in a memory. In some examples, the processor may be implemented as an application-specific integrated circuit (ASIC). In some examples, the processor may include one or more of a field programmable gate array (FPGA), a central processor unit (CPU), a graphic processor unit (GPU), a digital signal processor (DSP). In some examples, the processor may include a finite state machine (FSM). Other implementations are also possible.

(93) In some examples, the memory may be integrated with the processor. The memory may be a non-transitory computer-readable medium configured to store various software programs and/or multiple groups of instructions. The memory may be configured to store the instructions for implementing the various methods and processes provided in accordance with the various examples of this description. The memory may be implemented in any way known in the art.

(94) In the examples described herein, a relaxation oscillator architecture is described, where the frequency of the oscillator is dependent on electron mobility (μ) rather than R and C. Electron mobility is a single parameter rather than two parameters (such as R and C), and is more easily controlled for a given manufacturing process. The examples herein also describe a circuit that can compensate for any temperature coefficient of the manufacturing process. A mobility-based oscillator implementation as described herein improves the accuracy of the oscillator. The integration of the temperature compensation is inside the circuit to cancel the electron mobility variation across temperature. In examples herein, an external temperature sensor is not used for temperature compensation. Also, examples herein provide a reduction in test time, test cost, and circuit area for an oscillator circuit due to the elimination of trim bits for the oscillator.

(95) In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

(96) A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

(97) A circuit or device that is described herein as including certain components may instead be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

(98) While certain components may be described herein as being of a particular process technology, these components may be exchanged for components of other process technologies. Circuits described herein are reconfigurable to include the replaced components to provide functionality at

least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the shown resistor. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

(99) Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

(100) As used herein, the terms “terminal,” “node,” “interconnection,” “pin,” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device, or a semiconductor component. Furthermore, a voltage rail or more simply a “rail,” may also be referred to as a voltage terminal and may generally mean a common node or set of coupled nodes in a circuit at the same potential.

Claims

1. A system, comprising: a first transistor having a control terminal and first and second terminals, the first terminal coupled to a current mirror and the control terminal coupled to a first current source and a resistor; a second transistor having a control terminal and first and second terminals, the first terminal coupled to the current mirror, the second terminal coupled to the second terminal of the first transistor, and the control terminal coupled to the resistor and a second current source; a third transistor having a control terminal and first and second terminals, the first terminal coupled to a voltage terminal, the second terminal coupled to the control terminal of the second transistor, and the control terminal coupled to the first terminal of the second transistor; and a fourth transistor having a control terminal and first and second terminals, the control terminal coupled to the current mirror, the first and second terminals coupled to one another and to the second terminal of the first transistor and the second terminal of the second transistor.
2. The system of claim 1, wherein the first current source is coupled to the voltage terminal.
3. The system of claim 1, wherein the fourth transistor is configured as a metal-oxide semiconductor (MOS) capacitor.
4. The system of claim 1, further comprising: a first switch and a second switch coupled to the control terminal of the fourth transistor.
5. The system of claim 4, wherein the first switch is coupled between the control terminal of the fourth transistor and the current mirror, and the second switch is coupled between the control terminal of the fourth transistor and a second current mirror.
6. The system of claim 1, further comprising: a fifth transistor having a control terminal and first and second terminals, the first terminal coupled to the current mirror, the control terminal coupled to the first terminal, and the second terminal coupled to the second terminal of the first transistor and the second terminal of the second transistor; a sixth transistor having a control terminal and first and second terminals, the first terminal coupled to a second current mirror, and the second terminal coupled to the second terminal of the first transistor and the second terminal of the second transistor; and a divider coupled to the control terminal of the fifth transistor and to the control terminal of the sixth transistor.

7. The system of claim 6, wherein the fifth transistor and the sixth transistor are bipolar junction transistors.

8. The system of claim 1, wherein the second transistor is larger than the first transistor.

9. The system of claim 1, wherein the second current source is larger than the first current source.

10. A system, comprising: a first transistor having a control terminal and first and second terminals, the first terminal coupled to a current mirror and the control terminal coupled to a first current source and a resistor; a second transistor having a control terminal and first and second terminals, the first terminal coupled to the current mirror, the second terminal coupled to the second terminal of the first transistor, and the control terminal coupled to the resistor and a second current source; a third transistor having a control terminal and first and second terminals, the first terminal coupled to a voltage terminal, the second terminal coupled to the control terminal of the second transistor, and the control terminal coupled to the first terminal of the second transistor; a fourth transistor having a control terminal and first and second terminals, the control terminal coupled to the current mirror, the first and second terminals coupled to one another and to the second terminal of the first transistor and the second terminal of the second transistor; and a comparator having a comparator output and first and second comparator inputs, wherein the control terminal of the fourth transistor is coupled to the first comparator input.

11. The system of claim 10, further comprising: a first switch and a second switch coupled to the control terminal of the fourth transistor.

12. The system of claim 11, further comprising: a third switch and a fourth switch coupled to the second comparator input, wherein the third switch is coupled to a first voltage reference terminal and the fourth switch is coupled to a second voltage reference terminal.

13. The system of claim 12, further comprising: an inverter coupled to the comparator output.

14. The system of claim 10, further comprising: a fifth transistor having a control terminal and first and second terminals, the first terminal coupled to the current mirror, the control terminal coupled to the first terminal, and the second terminal coupled to the second terminal of the first transistor and the second terminal of the second transistor; a sixth transistor having a control terminal and first and second terminals, the first terminal coupled to a second current mirror, and the second terminal coupled to the second terminal of the first transistor and the second terminal of the second transistor; and a divider coupled to the control terminal of the fifth transistor and to the control terminal of the sixth transistor.

15. A system, comprising: a first transistor having a control terminal and first and second terminals, the first terminal coupled to a current mirror and the control terminal coupled to a first current source and a resistor, the first current source configured to provide a proportional to absolute temperature (PTAT) current; a second transistor having a control terminal and first and second terminals, the first terminal coupled to the current mirror, the second terminal coupled to the second terminal of the first transistor, and the control terminal coupled to the resistor and a second current source; a third transistor having a control terminal and first and second terminals, the first terminal coupled to a voltage terminal, the second terminal coupled to the control terminal of the second transistor, and the control terminal coupled to the first terminal of the second transistor; a fourth transistor having a control terminal and first and second terminals, the control terminal coupled to the current mirror, the first and second terminals coupled to one another and to the second terminal of the first transistor and the second terminal of the second transistor, the fourth transistor configured to receive a first current from the current mirror at its control terminal; and a comparator having a comparator output and first and second comparator inputs, wherein the control terminal of the fourth transistor is coupled to the first comparator input, and wherein the comparator is configured to compare a voltage at the control terminal of the fourth transistor to a reference voltage.

16. The system of claim 15, wherein the fourth transistor is further configured to provide an oscillating output signal at its control terminal.

17. The system of claim 16, further comprising: a first switch and a second switch coupled to the control terminal of the fourth transistor.
 18. The system of claim 17, further comprising: an inverter coupled to the comparator output.
 19. The system of claim 18, wherein the comparator is configured to provide a first output signal and the inverter is configured to provide a second output signal, wherein the first output signal is configured to control the first switch and the second output signal is configured to control the second switch.
 20. The system of claim 16, wherein a frequency of the oscillating output signal is dependent on electron mobility.
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